

# Linear Algebra for Optimization

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# What is this course about?

- **Goal:** build a **complete toolkit** to **model, analyze, and solve** optimization problems.
- **Foundations (why it works):**
  - ▶ numerical linear algebra for optimization;
  - ▶ convex geometry and convex analysis;
  - ▶ first-order optimality and constraint qualifications  $\Rightarrow$  **KKT** and certificates.
- **Duality as a unifying lens:** Lagrangians, strong duality (Slater), multipliers as **sensitivity prices**,  $\text{KKT} \Leftrightarrow$  saddle points (convex).
- **Algorithms (how to compute):**
  - ▶ gradient methods, steepest descent, conjugate gradient; rates and conditioning;
  - ▶ Newton and quasi-Newton (BFGS/L-BFGS): fast local accuracy with globalization;
  - ▶ constrained methods: a brief overview;
  - ▶ a quick look at stochastic gradient methods.
- **Throughout:** emphasize **certificates** (gaps, gradient mappings, dual bounds), **assumptions**, and **when guarantees do or do not apply**.

# Course structure (12 sessions)

- 1 Numerical linear algebra
- 2 Convex sets
- 3 Convex functions
- 4 Optimality conditions
- 5 Convex duality
- 6 Optimization zoo
- 7 Gradient descent
- 8 Newton's method
- 9 Constrained optimization
- 10 Interior-point methods
- 11 Stochastic gradient
- 12 Exam

# Why start with linear algebra?

- Your algorithms will be written in terms of **norms**, **inner products**, and **linear systems**.
- Convergence of Gradient Descent, Conjugate Gradient, Newton, Interior-Point Methods is essentially governed by **spectra**:  $\lambda_{\min}$ ,  $\lambda_{\max}$  and **conditioning**.
- In this course we will use mostly **symmetric matrices** (PSD order, spectral theorem), but we need Singular Value Decomposition (SVD) for operator norms and conditioning.
- Goal for today: fix a common vocabulary and a few key facts we will reuse all semester.

# Convention in these slides

## Icons / tags:

- ♥ : really important results
- ◇ : more advanced content
- ♣ : very simple in-class exercise
- ♠ : training exercise (harder)

BV x.y : *Convex Optimization*, S. Boyd,  
L. Vanderberghe, Ch. x, Sec. y

JCG x.y : *Fragments d'Optimisation  
Différentiable* (J.-Ch. Gilbert), Ch. x,  
Sec. y

- ➡ both are available online for free.
- ➡ Boyd's classes on convex optimization are on YouTube.

## Color coding<sup>a</sup>:

- In text:
  - ▶ violet for definitions,
  - ▶ red for key results/theorems/assumptions.
- In math:
  - ▶  $x$  primal variables,
  - ▶  $\lambda$  multipliers,
  - ▶  $\theta$  parameters,
  - ▶  $L$  smoothness constants,
  - ▶  $x^{(k)}$  iterates,
  - ▶  $\tau^{(k)}$  step sizes,
  - ▶  $d$  directions.

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<sup>a</sup>Best effort

# Contents

## 1 Symmetric matrices: spectral theorem, PSD order, projectors

- Diagonalization 101
- Spectral decomposition
- Projectors and PSD order

## 2 Inner products, norms, and dual norms

- Inner products and norms
- Gradient, Hessian, and Taylor expansion recap
- Dual norms and induced operator norms

## 3 Factorizations and solving linear systems

- Solving linear systems via factorizations
- LU and Cholesky factorizations
- QR factorization and least squares

## 4 SVD and conditioning

- Singular Value Decomposition (SVD)
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# Eigenvalues, eigenvectors, eigenspaces, spectrum (recall)



Let  $A \in \mathbb{R}^{n \times n}$  (we will always consider real matrices).

- A nonzero vector  $x \neq 0$  is an **eigenvector** of  $A$  associated with **eigenvalue**  $\lambda \in \mathbb{R}$  if

$$Ax = \lambda x.$$

- The **eigenspace** associated with  $\lambda$  is

$$E_\lambda(A) := \ker(A - \lambda I) = \{x \in \mathbb{R}^n : Ax = \lambda x\}.$$

- The **spectrum** of  $A$  is the set of its eigenvalues:

$$\text{sp}(A) := \{\lambda \in \mathbb{R} : \det(A - \lambda I) = 0\}.$$

♣ Exercise: Show that  $A$  is invertible iff  $0 \notin \text{sp}(A)$ .





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# Diagonalization (general case): definition and criterion

Let  $A \in \mathbb{R}^{n \times n}$ .

- $A$  is **diagonalizable** (over  $\mathbb{R}$ ) if there exist an invertible matrix  $P$  and a diagonal matrix  $D$  such that

$$A = PDP^{-1}.$$

- In that case, the diagonal entries of  $D$  are eigenvalues of  $A$ , and the columns of  $P$  form a basis of eigenvectors.

## Diagonalization criterion (via eigenspaces)

$A$  is diagonalizable over  $\mathbb{R}$  iff:

- 1 all eigenvalues are real (so  $\text{sp}(A) \subset \mathbb{R}$ ), and
- 2 the eigenvectors span  $\mathbb{R}^n$ , equivalently

$$n = \sum_{\lambda \in \text{sp}(A)} \dim(E_{\lambda}(A)).$$

♣ Exercise: Assume  $A$  has  $n$  *distinct* real eigenvalues. Show that  $A$  is diagonalizable.

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## (Semi)definite matrices and PSD order

Let  $S_n$  be the set of real symmetric  $n \times n$  matrices.

- **Positive semidefinite (PSD):**  $A \succeq 0$  (equivalently  $A \in S_n^+$ ) iff

$$x^\top A x \geq 0 \quad \forall x \in \mathbb{R}^n.$$

- **Positive definite (PD):**  $A \succ 0$  (equivalently  $A \in S_n^{++}$ ) iff

$$x^\top A x > 0 \quad \forall x \neq 0.$$

### Loewner (PSD) order on $S_n$

For  $A, B \in S_n$ , define

$$A \preceq B \iff B - A \succeq 0.$$

Equivalently,  $A \preceq B$  iff  $x^\top A x \leq x^\top B x$  for all  $x \in \mathbb{R}^n$ .

♣ Exercise: Show that  $\preceq$  is a partial order on  $S_n$  (reflexive, antisymmetric, transitive).

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# Spectral theorem (real symmetric matrices)



Let  $S_n$  be the set of real symmetric  $n \times n$  matrices.

## Theorem (Spectral theorem)

*Any symmetric matrix  $A \in S_n$  has  $n$  real eigenvalues (counted with multiplicity)  $\lambda_1, \dots, \lambda_n$  and an orthonormal basis of eigenvectors  $(q_i)_{i \in [n]}$  i.e., such that:*

$$Aq_i = \lambda_i q_i, \quad \text{and} \quad q_i^\top q_j = \delta_{ij}, \quad \forall i, j.$$

*In other words, there exists an orthogonal matrix  $Q$  (i.e.  $Q^\top Q = I$ ) such that*

$$A = Q \Lambda Q^\top,$$

*where  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$  and  $\lambda_1, \dots, \lambda_n$  are the eigenvalues of  $A$ .*



For  $A \in S_n$  with spectral decomposition  $A = Q\Lambda Q^\top$ , where  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ :

- $A \in S_n^+ \iff A \succeq 0 \iff \lambda_i \geq 0$  for all  $i$ .
- $A \in S_n^{++} \iff A \succ 0 \iff \lambda_i > 0$  for all  $i$ .
- For any  $A \in S_n^+$ , with  $A = Q\Lambda Q^\top$ , we have

$$A^{1/2} = Q\Lambda^{1/2}Q^\top, \quad \text{where} \quad \Lambda^{1/2} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}),$$

where  $A^{1/2}$  is the unique PSD square root of  $A$  (i.e.  $A^{1/2}A^{1/2} = A$ ).

♠ Exercise: Prove the above statements using the spectral decomposition.



# Rayleigh quotient and eigenvalue extrema



For  $A \in S_n$  and  $x \neq 0$ , define  $R_A(x) = \frac{x^\top A x}{x^\top x}$ .

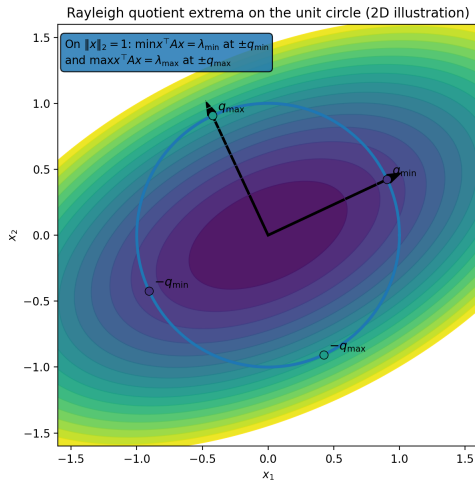
- $\lambda_{\min}(A) = \min_{\|x\|_2=1} x^\top A x,$

- $\lambda_{\max}(A) = \max_{\|x\|_2=1} x^\top A x.$

➔ min and max are attained at eigenvectors associated with  $\lambda_{\min}(A)$  and  $\lambda_{\max}(A)$ .

- If  $A \succ 0$ , then  $\|x\|_A^2 = x^\top A x$  satisfies

$$\lambda_{\min}(A) \|x\|_2^2 \leq \|x\|_A^2 \leq \lambda_{\max}(A) \|x\|_2^2.$$



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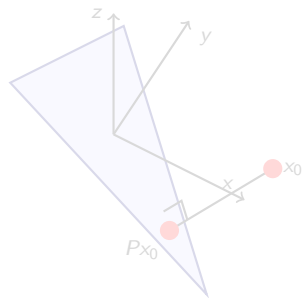
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# Orthogonal projectors

A matrix  $P$  is an **orthogonal projector** iff  $P^2 = P$  and  $P = P^\top$ .

- Then  $\mathbb{R}^n = \text{Im}(P) \oplus \ker(P)$ , and  $Px$  is the closest (for  $\|\cdot\|_2$ ) point to  $x$  in  $\text{Im}(P)$ .
- Eigenvalues of  $P$  are 0 or 1, so  $P \succeq 0$  and  $\|P\|_2 = 1$  (unless  $P = 0$ ).
- If  $\|q\|_2 = 1$ , then  $P = qq^\top$  is the orthogonal projector onto  $\text{span}(q)$ .
- More generally, if  $Q \in \mathbb{R}^{n \times k}$  has orthonormal columns, then  $P = QQ^\top$  projects onto  $\text{Im}(Q)$ .

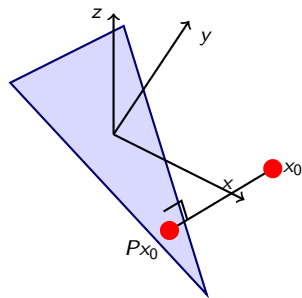


$Px_0$  is the Euclidean closest point to  $x_0$  in  $\text{Im}(P)$ .

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# Spectral decomposition as a sum of rank-one matrices

Let  $A \in S_n$  with spectral decomposition  $A = Q\Lambda Q^\top$ , where  $Q = [q_1 \cdots q_n]$  is orthogonal and  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ .

## Rank-one expansion

$$A = \sum_{i=1}^n \lambda_i q_i q_i^\top.$$

Each term  $q_i q_i^\top$  is the orthogonal projector onto  $\text{span}(q_i)$  (rank one).

Explanation: Since  $Q\Lambda Q^\top = \sum_{i=1}^n \lambda_i Q e_i e_i^\top Q^\top$  and  $Q e_i = q_i$ ,

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# PSD order and basic algebra

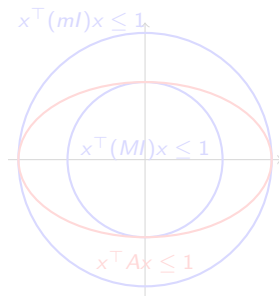
For  $A, B \in S_n$ , recall that  $A \preceq B$  iff  $B - A \succeq 0$ .

- Order is compatible with quadratic forms:  
 $A \preceq B \Rightarrow x^\top A x \leq x^\top B x$  for all  $x$ .
- If  $0 \prec A \preceq B$ , then  $B^{-1} \preceq A^{-1}$  (order reverses under inversion).

- $A \preceq B$  implies  $\lambda_{\min}(A) \leq \lambda_{\min}(B)$  and  $\lambda_{\max}(A) \leq \lambda_{\max}(B)$ .

→  $A \preceq MI$  iff  $\lambda_{\max}(A) \leq M$ , and  $A \succeq ml$  iff  $\lambda_{\min}(A) \geq m$ .

→ We will constantly use  $ml \preceq \nabla^2 f(x) \preceq MI$  to control GD and Newton.

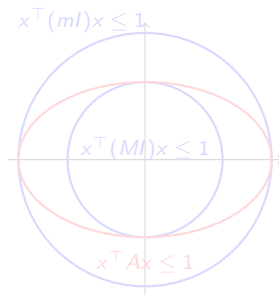


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- $A \preceq B$  implies  $\lambda_{\min}(A) \leq \lambda_{\min}(B)$  and  $\lambda_{\max}(A) \leq \lambda_{\max}(B)$ .
- ➔  $A \preceq M I$  iff  $\lambda_{\max}(A) \leq M$ , and  $A \succeq m I$  iff  $\lambda_{\min}(A) \geq m$ .

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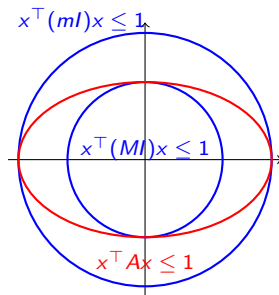




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A **norm** on  $\mathbb{R}^n$  is a map  $\| \cdot \| : \mathbb{R}^n \rightarrow \mathbb{R}_+$  such that:

- **Positive definiteness:**  $\|x\| \geq 0$  and  $\|x\| = 0 \iff x = 0$ .
- **Absolute homogeneity:**  $\|\alpha x\| = |\alpha| \|x\|$  for all  $\alpha \in \mathbb{R}$ .
- **Triangle inequality:**  $\|x + y\| \leq \|x\| + \|y\|$ .
- Classical examples:

Useful inequalities (for  $x \in \mathbb{R}^n$ ):

# Norms: a quick recall



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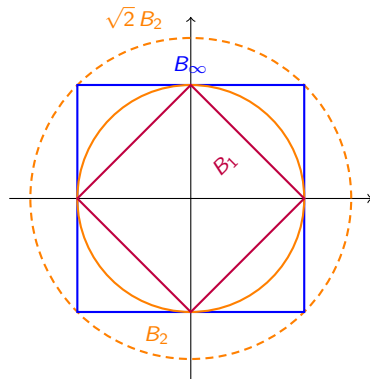
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- **Triangle inequality:**  $\|x + y\| \leq \|x\| + \|y\|$ .
- Classical examples:

$$\|x\|_2 = \sqrt{\sum_i x_i^2}, \quad \|x\|_1 = \sum_i |x_i|, \quad \|x\|_\infty = \max_i |x_i|.$$

- We say that two norms  $\|\cdot\|_a$  and  $\|\cdot\|_b$  are **equivalent** if there exist constants  $c, C > 0$  such that

$$c \|\cdot\|_a \leq \|\cdot\|_b \leq C \|\cdot\|_a.$$

- In **finite dimension**, all norms are equivalent!



Useful inequalities (for  $x \in \mathbb{R}^n$ ):

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2.$$

# Inner products and geometry

- An **inner product** on  $\mathbb{R}^n$  is a map  $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$  such that:
  - ▶ **Bilinear**:  $\langle ax_1 + bx_2, y \rangle = a\langle x_1, y \rangle + b\langle x_2, y \rangle$  and  $\langle x, ay_1 + by_2 \rangle = a\langle x, y_1 \rangle + b\langle x, y_2 \rangle$ .
  - ▶ **Symmetric**:  $\langle x, y \rangle = \langle y, x \rangle$ .
  - ▶ **Positive definite**:  $\langle x, x \rangle \geq 0$  and  $\langle x, x \rangle = 0 \iff x = 0$ .
- It defines a norm:  $\|x\| := \sqrt{\langle x, x \rangle}$ .
- **Cauchy–Schwarz**:  $|\langle x, y \rangle| \leq \|x\| \|y\|$ .
  - ▶ **Equality case**: if  $x \neq 0$  and  $y \neq 0$ , equality holds iff  $y = \alpha x$  for some  $\alpha \in \mathbb{R}$  (colinearity).
  - ▶ (Trivial cases: if  $x = 0$  or  $y = 0$ , equality holds with both sides = 0.)
- Orthogonality depends on the inner product:  $x \perp y \iff \langle x, y \rangle = 0$ .

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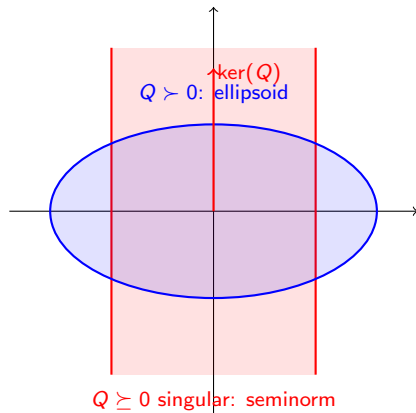
# The $\|\cdot\|_Q$ norm (and when it is not a norm)



Let  $Q \in S_n^+$ .

- Define  $\|x\|_Q := \sqrt{x^\top Q x}$ .
- If  $Q \succ 0$ :  $\|\cdot\|_Q$  is a **norm** and its unit ball is an **ellipsoid**.
- If  $Q \succeq 0$  but singular:  $\|\cdot\|_Q$  is a **seminorm**, with nullspace  $\ker(Q) = \{x : Qx = 0\}$ .
- Change of variables (SPD case):  $\|x\|_Q = \|Q^{1/2}x\|_2$ .

♠ Exercise: Show  $\|x\|_Q \leq \sqrt{\lambda_{\max}(Q)} \|x\|_2$  and  $\|x\|_2 \leq \frac{1}{\sqrt{\lambda_{\min}(Q)}} \|x\|_Q$  when  $Q \succ 0$ .



Unit “ball”: ellipse (SPD) vs unbounded strip (singular PSD).



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## Gradient (recap)



Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be differentiable at  $x$ .

Gradient = the unique vector defining the first-order approximation

There exists a **unique** vector  $g \in \mathbb{R}^n$  such that

$$f(x + h) = f(x) + g^\top h + o(\|h\|_2) \quad \text{as } h \rightarrow 0.$$

We denote it  $g = \nabla f(x)$ .

### Coordinate formula

In the canonical basis,

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}.$$

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# Hessian and second-order Taylor expansion (recap)



If  $f$  is twice differentiable at  $x$ :

Hessian = the unique matrix defining the second-order approximation

There exists a **unique** symmetric matrix  $H \in S_n$  such that

$$f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H h + o(\|h\|_2^2) \quad \text{as } h \rightarrow 0.$$

We denote it  $H = \nabla^2 f(x)$ .

Coordinate formula

$$\nabla^2 f(x) = \left[ \frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right]_{i,j=1}^n.$$

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# Hessian bounds and strong convexity



Assume  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is twice differentiable.

## PSD order $\iff$ eigenvalue bounds

For any symmetric matrix  $H \in S_n$  and any  $m \in \mathbb{R}$ ,

$$H \succeq mI \iff \lambda_{\min}(H) \geq m \iff x^\top Hx \geq m\|x\|_2^2 \quad \forall x \in \mathbb{R}^n.$$

## Strong convexity via the Hessian

If there exists  $m > 0$  such that

$$\nabla^2 f(x) \succeq mI \quad \forall x \in \mathbb{R}^n,$$

then  $f$  is  $m$ -strongly convex (w.r.t.  $\|\cdot\|_2$ ), i.e.

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|_2^2 \quad \forall x, y \in \mathbb{R}^n.$$

Equivalently,  $\lambda_{\min}(\nabla^2 f(x)) \geq m$  for all  $x$ .

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# Dual norm: definition and two key examples

Given a norm  $\|\cdot\|$ , its **dual norm** is

$$\|y\|_{\star} := \sup_{\|x\| \leq 1} y^{\top} x.$$

- Generalized Cauchy-Schwarz:  $y^{\top} x \leq \|y\|_{\star} \|x\|$ .
- Examples:
  - ▶  $(\|\cdot\|_2)^{\star} = \|\cdot\|_2$ ,
  - ▶  $(\|\cdot\|_1)^{\star} = \|\cdot\|_{\infty}$ ,
  - ▶  $(\|\cdot\|_{\infty})^{\star} = \|\cdot\|_1$ .



For matrix  $A$ , the **operator norm induced** by  $\|\cdot\|_{op}$  is

$$\|A\|_{op} := \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|}.$$

- For  $\|\cdot\|_2$ :  $\|A\|_{2,op}$  is the **spectral norm**.
- Equivalent characterizations:
  - ▶  $\|A\|_{2,op} = \sqrt{\lambda_{\max}(A^\top A)}$ .
  - ▶  $\|A\|_{2,op} = \sigma_{\max}(A)$  (largest singular value – see later).
- For  $A \in S_n$ :  $\|A\|_{2,op} = \max_i |\lambda_i(A)|$
- For  $A \in S_n^+$ ,  $\|A\|_{2,op} = \lambda_{\max}(A)$ .

# Frobenius norm (matrix “Euclidean” norm)



For  $A \in \mathbb{R}^{m \times n}$ , define

$$\|A\|_F := \sqrt{\sum_{i,j} A_{ij}^2} = \sqrt{\text{Tr}(A^\top A)}.$$

If  $A = U\Sigma V^\top$  is an SVD with singular values  $(\sigma_i)$ , then

$$\|A\|_F^2 = \sum_{i=1}^p \sigma_i^2, \quad \|A\|_2 \leq \|A\|_F \leq \sqrt{\text{rank}(A)} \|A\|_2.$$

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## Preview: Newton as repeated quadratic minimization

Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be twice differentiable. Use second order Taylor expansion around the current iterate  $x^{(k)}$ :

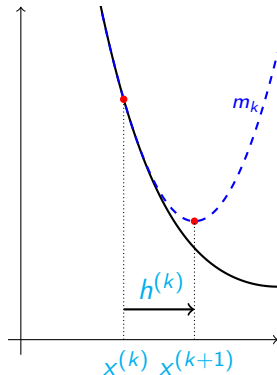
$$\begin{aligned} f(x^{(k)} + h) &= m_k(h) + o(\|h\|_2^2), \\ m_k(h) &:= f(x^{(k)}) + \nabla f(x^{(k)})^\top h \\ &\quad + \frac{1}{2} h^\top \nabla^2 f(x^{(k)}) h. \end{aligned}$$

### Newton idea

Newton<sup>a</sup> is: build a quadratic model, minimize it, repeat.

$$\begin{aligned} h^{(k)} &\leftarrow \arg \min_h m_k(h) \\ x^{(k+1)} &= x^{(k)} + h^{(k)} \end{aligned}$$

<sup>a</sup>Often the fastest *local* method under strong regularity assumptions.



Newton step as quadratic model minimization

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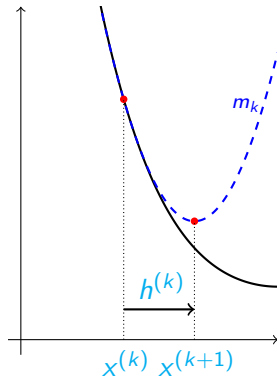
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Newton step as quadratic model minimization

## Minimizing the quadratic model gives a linear system

Recall the local quadratic model at  $x^{(k)}$ :

$$m_k(h) := f(x^{(k)}) + \nabla f(x^{(k)})^\top h + \frac{1}{2} h^\top \nabla^2 f(x^{(k)}) h.$$

### Optimality condition

A minimizer  $h^{(k)}$  satisfies  $\nabla m_k(h^{(k)}) = 0$ . Since

$$\nabla m_k(h) = \nabla f(x^{(k)}) + \nabla^2 f(x^{(k)}) h,$$

we obtain the (symmetric) linear system

$$\nabla^2 f(x^{(k)}) h^{(k)} = -\nabla f(x^{(k)}).$$

If  $\nabla^2 f(x^{(k)}) \succ 0$ , the minimizer is unique.

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# How to solve $Ax = b$ ?



Goal: solve one linear system  $Ax = b$ .

## Triangular systems are easy

If  $L$  is lower triangular,  $Lx = b$  is solved by **forward substitution** (one pass).

If  $U$  is upper triangular,  $Ux = b$  is solved by **backward substitution** (one pass).

$\leadsto O(n^2)$  operations each.

## Computing $A^{-1}$ means solving $n$ systems

Let  $e_1, \dots, e_n$  be the canonical basis. Then

$$A^{-1} = [x^{(1)} \ \dots \ x^{(n)}] \quad \text{with} \quad Ax^{(i)} = e_i \quad (i = 1, \dots, n).$$

So forming  $A^{-1}$  amounts to solving  $n$  right-hand sides.

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**Conclusion: solve  $Ax = b$  with dedicated tools, do not invert.**

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**Conclusion: solve  $Ax = b$  with dedicated tools, do not invert.**

# Why factorize?

In optimization, we repeatedly solve linear systems  $Ax = b$  (Newton steps, least-squares, KKT blocks, ...).

## Principle

Factorize once, solve many times.

- We reuse the same factorization, we do **not** compute  $A^{-1}$ .
- The expensive part is the factorization; each new right-hand side is cheap (triangular solves).

♣ Exercise: Give two different vectors  $b^{(1)}, b^{(2)}$ . Explain why reusing the same factorization of  $A$  is efficient.

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## LU factorization (square matrices)

For a general square matrix  $A \in \mathbb{R}^{n \times n}$ , we aim for

$$PA = LU,$$

where  $L$  is lower triangular (often with ones on the diagonal) and  $U$  is upper triangular, and  $P$  is a permutation matrix (pivoting for numerical stability).

♣ Exercise: Assume  $A = LU$ . Show that solving  $Ax = b$  reduces to two triangular solves. Show that it stays the same if  $PA = LU$ .



If  $A \in S_n^{++}$  (i.e.,  $A \succ 0$ ), then there exists a unique lower triangular  $L$  with positive diagonal such that

$$A = LL^\top.$$

- Solve  $Ax = b$  via:

$$Ly = b, \quad L^\top x = y.$$

- This is the standard path for Hessian systems in strongly convex optimization.

♣ Exercise: Show that if  $A = LL^\top$  and  $L$  is invertible, then  $A \succ 0$ .

# Why $LL^T$ (Cholesky) over $LU$ ?



Assume  $A \in S_n$  and  $A \succ 0$ .

- **Faster (dense):** Cholesky costs about  $\frac{1}{3}n^3$  flops vs about  $\frac{2}{3}n^3$  for LU.
- **Less memory:** store essentially one triangular factor (symmetry is exploited).
- **More robust here:** for SPD matrices, Cholesky does not require pivoting.
- **Matches optimization geometry:**  $x^T Ax = \|L^T x\|_2^2$ .



## Worked example: Cholesky on a $2 \times 2$ SPD matrix



Let

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

We look for  $A = LL^T$  with  $L = \begin{pmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{pmatrix}$ .

Matching entries:

$$\ell_{11} = 2, \quad \ell_{21} = 1, \quad \ell_{22} = \sqrt{2}, \quad \Rightarrow \quad L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}.$$

Solve  $Ly = b$ :  $y_1 = 1$ ,  $y_2 = 0$ . Then solve  $L^T x = y$ :  $x_2 = 0$ ,  $x_1 = \frac{1}{2}$ .

$$x = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}.$$

♣ Exercise: Change  $b$  to  $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ . Reuse the same  $L$  and solve again.

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## Takeaways (what to remember)

- Solving  $Ax = b$  is done via **factorization + triangular solves**, not via  $A^{-1}$ .
- Use **Cholesky** when  $A$  is symmetric positive definite:  $A = LL^\top$ .
- Use **LU** as the general-purpose tool (often with a permutation  $PA = LU$ ).
- Reuse factorizations whenever the matrix  $A$  stays the same and only  $b$  changes.

♠ Exercise: (Optimization link) Consider  $f(x) = \frac{1}{2}x^\top Ax - b^\top x$  with  $A \succ 0$ . Show that the minimizer solves  $Ax = b$ , and explain how Cholesky gives the minimizer efficiently.

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## QR factorization (thin QR)

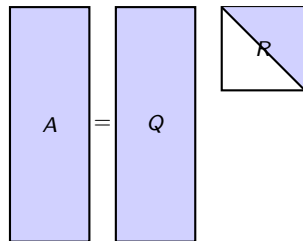


For a matrix  $A \in \mathbb{R}^{m \times n}$ , with  $m \geq n$ , there exist matrices  $Q \in \mathbb{R}^{m \times n}$  and  $R \in \mathbb{R}^{n \times n}$  such that

$$A = QR,$$

$$Q^\top Q = I_n,$$

$R$  upper triangular.



In particular:

- $Q$  has **orthonormal columns**. If  $\text{rank}(A) = n$ , they form a basis of  $\text{Im}(A)$ .
- $R$  is a **change of coordinates** in that basis (triangular  $\Rightarrow$  easy solves).
- If  $A$  has full column rank, then  $R$  is **invertible**.
- Note that  $P := QQ^\top$  is the orthogonal projector onto  $\text{Im}(Q)$ .

tall matrix  $m \geq n$

Blue = potentially nonzero entries .

# Least squares and why we avoid normal equations



Given  $A \in \mathbb{R}^{m \times n}$  and  $b \in \mathbb{R}^m$ , the (linear) least-squares problem is

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_2^2.$$

## Optimality condition (normal equations)

If  $A$  has full column rank, the objective is differentiable and strictly convex, and the unique minimizer  $x^*$  satisfies

$$A^\top(Ax^* - b) = 0 \quad \Longleftrightarrow \quad A^\top A x^* = A^\top b.$$

## But: normal equations can be numerically bad

Forming  $A^\top A$  typically **squares the conditioning**:

$$\kappa_2(A^\top A) = \kappa_2(A)^2 \quad (\text{when } A \text{ has full column rank}).$$

**Conclusion:** solve LS via **QR** (or SVD), not via  $A^\top A$ .

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## Solving least squares via QR (thin QR)



Assume  $m \geq n$ . Let  $A = QR$  be a thin QR factorization with

$$Q \in \mathbb{R}^{m \times n}, \quad Q^\top Q = I_n, \quad R \in \mathbb{R}^{n \times n} \text{ upper triangular.}$$

### Key step: orthogonal decomposition

Let  $P := QQ^\top$  (orthogonal projector onto  $\text{Im}(Q)$ ). Then for any  $x$ ,

$$\|Ax - b\|_2^2 = \|QRx - b\|_2^2 = \|Rx - Q^\top b\|_2^2 + \|(I - P)b\|_2^2.$$

### Consequence (full column rank)

If  $A$  has full column rank, then  $R$  is invertible and the unique minimizer satisfies

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solved by backward substitution.

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# Singular Value Decomposition (SVD): definition



SVD extends spectral decomposition to **rectangular** matrices.

## Existence (main requirement)

For any  $A \in \mathbb{R}^{m \times n}$ , there exist orthogonal matrices  $U \in \mathbb{R}^{m \times m}$ ,  $V \in \mathbb{R}^{n \times n}$ , and a **rectangular diagonal** matrix  $\Sigma \in \mathbb{R}^{m \times n}$  such that

$$A = U\Sigma V^T, \quad \Sigma = \begin{cases} \text{diag}(\sigma_1, \dots, \sigma_n) & \text{if } m \geq n, \\ \text{diag}(\sigma_1, \dots, \sigma_m) & \text{if } m < n, \end{cases}$$

with  $p = \min(m, n)$  and  $\sigma_1 \geq \dots \geq \sigma_p \geq 0$ .

- $U, V$ : orthogonal changes of basis (rotations/reflections).
- $\Sigma$ : axis scalings; zeros encode directions collapsed by  $A$ .
- If  $r = \text{rank}(A)$ :  $\sigma_1 \geq \dots \geq \sigma_r > 0$  and  $\sigma_{r+1} = \dots = \sigma_p = 0$ .
- Geometry:  $A(B_2)$  is an ellipsoid with semi-axes  $\sigma_i$  (directions = columns of  $U$ ).

# Singular Value Decomposition (SVD): definition



SVD extends spectral decomposition to **rectangular** matrices.

## Existence (main requirement)

For any  $A \in \mathbb{R}^{m \times n}$ , there exist orthogonal matrices  $U \in \mathbb{R}^{m \times m}$ ,  $V \in \mathbb{R}^{n \times n}$ , and a **rectangular diagonal** matrix  $\Sigma \in \mathbb{R}^{m \times n}$  such that

$$A = U\Sigma V^T, \quad \Sigma = \begin{cases} \text{diag}(\sigma_1, \dots, \sigma_n) & \text{if } m \geq n, \\ \text{diag}(\sigma_1, \dots, \sigma_m) & \text{if } m < n, \end{cases}$$

with  $p = \min(m, n)$  and  $\sigma_1 \geq \dots \geq \sigma_p \geq 0$ .

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Let  $A = U\Sigma V^\top$  be an SVD with singular values  $\sigma_1 \geq \dots \geq \sigma_p$ .

- $\sigma_i^2$  are the positive eigenvalues of  $A^\top A$  (and of  $AA^\top$ ).
- $\|A\|_2 = \sigma_1$  (spectral/operator norm).

- 

$$\min_{\|x\|_2=1} \|Ax\|_2 = \begin{cases} \sigma_p & \text{if } \ker(A) = \{0\} \text{ (full column rank),} \\ 0 & \text{otherwise.} \end{cases}$$

- If  $A$  is square invertible:  $\|A^{-1}\|_2 = 1/\sigma_n$  and  $\kappa_2(A) = \sigma_1/\sigma_n$ .
- The truncated SVD gives the best rank- $k$  approximation of  $A$  in spectral and Frobenius norms.

# Contents

## 1 Symmetric matrices: spectral theorem, PSD order, projectors

- Diagonalization 101
- Spectral decomposition
- Projectors and PSD order

## 2 Inner products, norms, and dual norms

- Inner products and norms
- Gradient, Hessian, and Taylor expansion recap
- Dual norms and induced operator norms

## 3 Factorizations and solving linear systems

- Solving linear systems via factorizations
- LU and Cholesky factorizations
- QR factorization and least squares

## 4 SVD and conditioning

- Singular Value Decomposition (SVD)
- Condition numbers



- For  $A$  invertible,  $\kappa_2(A) := \|A\|_2 \|A^{-1}\|_2 = \sigma_1/\sigma_n \geq 1$ .
- For  $A \in S_n^{++}$ :  $\kappa_2(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ .
- Linear systems: relative error can be amplified by  $\kappa_2(A)$ .
- GD on quadratics: rates depend on  $\kappa$ .

Preview: CG improves  $\kappa$  to  $\sqrt{\kappa}$  in the convergence factor.

# Stability vs conditioning (linear solves)



Two different notions:

- **Conditioning** = sensitivity of the *problem*.

$Ax = b \Rightarrow$  how much does  $x$  change if  $b$  changes a bit?

- **Stability** = behavior of the *algorithm* in finite precision.

does the computed  $\hat{x}$  solve a nearby linear system?

## Rule of thumb

Even with a very good algorithm, the best relative accuracy you can expect is often on the order of

$$\frac{\|\hat{x} - x\|}{\|x\|} \approx O(\kappa_2(A) u),$$

where  $u$  is machine precision.

♣ Exercise: Take  $A = \text{diag}(1, \varepsilon)$ ,  $b = (1, 1)$ . What happens to the solution when  $\varepsilon$  is very small? Interpret with  $\kappa_2(A)$ .

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# Backward stability (the ideal guarantee)



## Definition (informal)

An algorithm for solving  $Ax = b$  is **backward stable** if it returns  $\hat{x}$  such that

$$(A + \Delta A)\hat{x} = b \quad \text{with} \quad \frac{\|\Delta A\|}{\|A\|} \text{ small (typically } O(u)\text{)}.$$

- Meaning: the algorithm behaves as if it solved *exactly* a slightly perturbed problem.
- If the problem is well-conditioned, a small backward error implies a small forward error.

## Practical takeaway

Good solvers are designed to be (close to) backward stable:

- SPD  $\Rightarrow$  Cholesky + triangular solves.
- General  $A \Rightarrow$  LU with pivoting.
- Least-squares  $\Rightarrow$  QR (avoid normal equations).

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## What you have to know

- Definitions: eigenvalues/eigenvectors; spectrum; PSD/PD; Loewner order  $A \preceq B$ .
- How to read  $mI \preceq A \preceq MI$  as a “geometry sandwich” (ellipsoid between balls) and as eigenvalue bounds.
- Spectral theorem for symmetric matrices:  $A = Q\Lambda Q^\top$  and consequences ( $A \succeq 0 \Leftrightarrow \lambda_i \geq 0$ ).
- Dual norms and induced operator norm; in particular  $\|A\|_2 = \sigma_{\max}(A)$  and  $\|A\|_2 = \max_i |\lambda_i(A)|$  for  $A \in S_n$ .

## What you really should know

- Rayleigh quotient facts:  $\lambda_{\min} = \min_{\|x\|_2=1} x^\top A x$ ,  $\lambda_{\max} = \max_{\|x\|_2=1} x^\top A x$ .
- The projector viewpoint:  $P^2 = P$ ,  $P = P^\top$ ,  $Px$  is the Euclidean closest point in  $\text{Im}(P)$ ;  $P = QQ^\top$ .
- Conditioning as the main complexity parameter:  $\kappa_2(A) = \lambda_{\max}/\lambda_{\min}$  (SPD), impacts GD/CG/Newton and numerical accuracy.
- Stability vs conditioning: good solvers aim for (near) backward stability; forward error typically scales like  $\kappa \cdot u$ .

## What you have to be able to do

- Use spectral decomposition to prove PSD/PD statements and derive inequalities like  $\lambda_{\min} \|x\|_2^2 \leq x^\top A x \leq \lambda_{\max} \|x\|_2^2$ .
- Recognize when  $\|\cdot\|_Q$  is a norm vs a seminorm; use  $\|x\|_Q = \|Q^{1/2}x\|_2$  when  $Q \succ 0$ .
- Solve  $Ax = b$  efficiently via factorizations (LU / Cholesky): factorize once, then triangular solves for new right-hand sides.

# What you should be able to do

- Diagnose ill-posedness: interpret tiny singular values / large  $\kappa$  as instability risk; know when SVD is the right diagnostic tool.
- Give quick “method choices” for linear algebra primitives: SPD  $\rightarrow$  Cholesky; general square  $\rightarrow$  pivoted LU; LS  $\rightarrow$  QR (or SVD).